Double Asymptotic Variance and Bias of the K-fold ELPPD Estimator under High-Dimensional Bernstein-von Mises

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Abstract

We establish the double asymptotic theory ($S \to \infty$ followed by $n \to \infty$) of the self-normalized importance sampling (SNIS) estimator of the K-fold Expected Log Pointwise Predictive Density (ELPPD) computed from Markov chain Monte Carlo (MCMC) samples. Under a high-dimensional Bernstein-von Mises (BvM) framework and polynomial ergodicity conditions on the MCMC sampler, we derive the exact scaling of the asymptotic variance and the bias of its estimator. We prove that the first-order MCMC fluctuations vanish due to the MLE score equations, and the leading-order variance is determined by a quadratic form of the MCMC draws. The resulting asymptotic variance scales as $\frac{2d(n)}{(1-1/K(n))^2}\gamma_{\text{mcmc}}$, where $d(n)$ is the parameter dimension and γ_{mcmc} is the MCMC efficiency factor.

1 Introduction and Notation

Let $y^{(n)} = (y_1, \ldots, y_n)^\top \in \mathcal{Y}^n$ be a vector of n independent observations drawn from the true data-generating distribution P_{θ_0} with density $p(y \mid \theta_0)$, where $\theta_0 \in \text{int}(\Theta)$ and $\Theta \subset \mathbb{R}^{d(n)}$ is the parameter space. The parameter dimension $d(n) = \dim(\theta)$ grows as a function of the sample size n. Let $\pi(\theta)$ be the prior distribution on Θ.

Let $\kappa : \{1, \ldots, n\} \to \{1, \ldots, K(n)\}$ define the fold partition. The training set for fold $k = \kappa(i)$ is denoted by $\mathcal{D}_{-k}$, which has size $n - n_k = n(1-1/K(n))$, where $n_k \approx n/K(n)$ is the size of the omitted fold. The fold-specific posterior distribution is denoted by $p(\theta \mid y_{\mathcal{D}_{-\kappa(i)}})$.

Let $\{\theta^{(s)}\}_{s=1}^S$ be a Markov chain with stationary distribution $q(\theta)$, serving as the proposal distribution. For each observation $i \in \{1, \ldots, n\}$, the unnormalized importance weights and weighted likelihoods are defined as:

$$w_{\kappa(i)}(\theta) = \frac{p(\theta \mid y_{\mathcal{D}_{-\kappa(i)}})}{q(\theta)}, \quad u_i(\theta) = p(y_i \mid \theta) w_{\kappa(i)}(\theta) \tag{1}$$

The true fold-specific posterior predictive density is:

$$\mu_i := p(y_i \mid y_{\mathcal{D}_{-\kappa(i)}}) = \int p(y_i \mid \theta) p(\theta \mid y_{\mathcal{D}_{-\kappa(i)}}) d\theta = \frac{\mathbb{E}_q[u_i(\theta)]}{\mathbb{E}_q[w_{\kappa(i)}(\theta)]} \tag{2}$$

and its self-normalized importance sampling (SNIS) estimator is:

$$\widehat{\mu}_i = \frac{\frac{1}{S}\sum_{s=1}^{S} u_i(\theta^{(s)})}{\frac{1}{S}\sum_{s=1}^{S} w_{\kappa(i)}(\theta^{(s)})} \tag{3}$$

The estimator of the K-fold Expected Log Pointwise Predictive Density (ELPPD) is:

$$\widehat{\text{ELPPD}} = \sum_{i=1}^{n} \log \widehat{\mu}_i \tag{4}$$

while the true mathematical K-fold ELPPD is $\text{ELPPD} = \sum_{i=1}^{n} \log \mu_i$.

2 High-Dimensional BvM CLT for Fold-Specific Estimators

We establish the joint asymptotic distribution of the fold-specific estimators as $n \to \infty$.

Assumption 1 (Regularity of Likelihood and Prior)**.** *We assume standard regularity conditions for growing parameter dimensions $d(n) \to \infty$ as $n \to \infty$ [3, 6]:*

1. *The dimension satisfies $d(n)^2/n \to 0$ as $n \to \infty$.*
2. *The log-likelihood function is twice continuously differentiable, and there exists a function $L(y)$ such that:*

$$\left\| \nabla_\theta^2 \log p(y \mid \theta) - \nabla_\theta^2 \log p(y \mid \theta_0) \right\|_{op} \leq L(y) \|\theta - \theta_0\|_2$$

   where $\mathbb{E}_{\theta_0}[L(Y)^2] \leq C_L < \infty$.
3. *The Fisher information matrix $I(\theta_0) = -\mathbb{E}_{\theta_0}[\nabla_\theta^2 \log p(Y_1 \mid \theta_0)]$ is positive definite with eigenvalues bounded: $c_0 I_d \leq I(\theta_0) \leq C_0 I_d$.*

Lemma 1 (Joint BvM CLT for Fold-Specific Estimators)**.** *Under Assumption 1, the fold-specific maximum likelihood estimators $\hat{\theta}_{-k}$ for $k \in \{1, \ldots, K\}$ satisfy:*

$$\sqrt{n}\big(\hat{\theta}_{-k} - \theta_0\big) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{1 - 1/K(n)} I(\theta_0)^{-1}\right) \quad \text{as } n \to \infty$$

and the covariance between estimators for different folds $k \neq l$ is:

$$Cov\left(\hat{\theta}_{-k}, \hat{\theta}_{-l}\right) = \frac{1}{n}\left(1 - \frac{n_k + n_l}{n}\right) I(\theta_0)^{-1} + o(1/n)$$

Proof. By standard asymptotic expansions for M-estimators in growing dimensions [7], the training MLE $\hat{\theta}_{-k}$ satisfies:

$$\hat{\theta}_{-k} - \theta_0 = I(\theta_0)^{-1} \frac{1}{n - n_k} \sum_{j \in \mathcal{D}_{-k}} \psi_j + o_p(1/\sqrt{n})$$

where $\psi_j = \nabla_\theta \log p(y_j \mid \theta_0)$ are independent with $\mathbb{E}[\psi_j] = 0$ and $\mathrm{Var}(\psi_j) = I(\theta_0)$. The variance of the sum is:

$$\mathrm{Var}\left(\hat{\theta}_{-k} - \theta_0\right) = I(\theta_0)^{-1} \left(\frac{1}{(n-n_k)^2} \sum_{j \in \mathcal{D}_{-k}} \mathrm{Var}(\psi_j) \right) I(\theta_0)^{-1} = \frac{1}{n-n_k} I(\theta_0)^{-1} = \frac{1}{n(1-1/K)} I(\theta_0)^{-1}$$

For two different folds $k \neq l$, the training sets are $\mathcal{D}_{-k}$ and $\mathcal{D}_{-l}$. The intersection of the training sets contains $n - n_k - n_l$ observations. The covariance is:

$$\mathrm{Cov}\left(\hat{\theta}_{-k}, \hat{\theta}_{-l}\right) = I(\theta_0)^{-1} \left(\frac{1}{(n-n_k)(n-n_l)} \sum_{j \in \mathcal{D}_{-k} \cap \mathcal{D}_{-l}} \mathrm{Var}(\psi_j) \right) I(\theta_0)^{-1} = \frac{n-n_k-n_l}{(n-n_k)(n-n_l)} I(\theta_0)^{-1}$$

Using the Taylor expansion of the scalar factor:

$$\frac{n-n_k-n_l}{(n-n_k)(n-n_l)} = \frac{1}{n} \frac{1-\frac{n_k+n_l}{n}}{(1-\frac{n_k}{n})(1-\frac{n_l}{n})} = \frac{1}{n}\left(1 - \frac{n_k+n_l}{n} + o(1)\right)$$

This yields the covariance formula. □

3 BvM Expansion of the Pointwise Predictive Density

We derive the asymptotic form of the posterior predictive density μ_i under BvM.

Theorem 1 (Pointwise Predictive Density Expansion)**.** *Under Assumption 1, the logarithm of the fold-specific posterior predictive density satisfies:*

$$\log \mu_i = \log p(y_i \mid \hat{\theta}_{-\kappa(i)}) + \frac{1}{2n(1-1/K)} tr\left[\left(\nabla_\theta^2 \log p(y_i \mid \theta_0) + \nabla_\theta \log p(y_i \mid \theta_0) \nabla_\theta \log p(y_i \mid \theta_0)^\top\right) I(\theta_0)^{-1}\right] + o_l$$

where $\hat{\theta}_{-\kappa(i)}$ is the MLE of the training set.

Proof. The predictive density is $\mu_i = \int p(y_i \mid \theta) p(\theta \mid y_{\mathcal{D}_{-\kappa(i)}}) d\theta$. By the high-dimensional BvM theorem, we replace the training posterior $p(\theta \mid y_{\mathcal{D}_{-\kappa(i)}})$ with its Gaussian approximation $\mathcal{N}(\hat{\theta}_{-\kappa(i)}, \Sigma_{-\kappa(i)})$, where $\Sigma_{-\kappa(i)} = \frac{1}{n(1-1/K)} I(\theta_0)^{-1}$.

We Taylor-expand $p(y_i \mid \theta)$ around the training MLE $\hat{\theta}_{-\kappa(i)}$:

$$p(y_i \mid \theta) = p(y_i \mid \hat{\theta}_{-\kappa(i)}) \exp\left(u^\top(\theta - \hat{\theta}_{-\kappa(i)}) + \frac{1}{2}(\theta - \hat{\theta}_{-\kappa(i)})^\top M (\theta - \hat{\theta}_{-\kappa(i)}) + o_p(\|\theta - \hat{\theta}_{-\kappa(i)}\|^2) \right)$$

where $u = \nabla_\theta \log p(y_i \mid \hat{\theta}_{-\kappa(i)})$ and $M = \nabla_\theta^2 \log p(y_i \mid \hat{\theta}_{-\kappa(i)})$.

Let $z = \theta - \hat{\theta}_{-\kappa(i)} \sim \mathcal{N}(0, \Sigma_{-\kappa(i)})$. We evaluate the expectation:

$$\mathbb{E}\left[\exp\left(u^\top z + \frac{1}{2} z^\top M z\right)\right] = \frac{1}{(2\pi)^{d/2} |\Sigma_{-\kappa(i)}|^{1/2}} \int \exp\left(u^\top z - \frac{1}{2} z^\top (\Sigma_{-\kappa(i)}^{-1} - M) z\right) dz$$

Using the standard multivariate Gaussian integral identity:

$$\int \exp\left(u^\top z - \frac{1}{2} z^\top A z\right) dz = (2\pi)^{d/2} |A|^{-1/2} \exp\left(\frac{1}{2} u^\top A^{-1} u\right)$$

where $A = \Sigma_{-\kappa(i)}^{-1} - M$. Thus:

$$\mathbb{E}\left[\exp\left(u^\top z + \frac{1}{2} z^\top M z\right)\right] = |I - \Sigma_{-\kappa(i)} M|^{-1/2} \exp\left(\frac{1}{2} u^\top (I - \Sigma_{-\kappa(i)} M)^{-1} \Sigma_{-\kappa(i)} u\right)$$

Since $\|\Sigma_{-\kappa(i)}\|_{\text{op}} = \mathcal{O}(1/n)$, we expand the determinant and matrix inverse:

$$|I - \Sigma_{-\kappa(i)} M|^{-1/2} = \exp\left(-\frac{1}{2} \log |I - \Sigma_{-\kappa(i)} M|\right) \approx \exp\left(\frac{1}{2} \text{tr}(\Sigma_{-\kappa(i)} M)\right) \approx 1 + \frac{1}{2} \text{tr}(\Sigma_{-\kappa(i)} M)$$

$$\exp\left(\frac{1}{2} u^\top (I - \Sigma_{-\kappa(i)} M)^{-1} \Sigma_{-\kappa(i)} u\right) \approx \exp\left(\frac{1}{2} u^\top \Sigma_{-\kappa(i)} u\right) \approx 1 + \frac{1}{2} u^\top \Sigma_{-\kappa(i)} u$$

Multiplying these expansions yields:

$$\mu_i \approx p(y_i \mid \hat{\theta}_{-\kappa(i)}) \left(1 + \frac{1}{2} \text{tr}(\Sigma_{-\kappa(i)} M) + \frac{1}{2} u^\top \Sigma_{-\kappa(i)} u\right) = p(y_i \mid \hat{\theta}_{-\kappa(i)}) \left(1 + \frac{1}{2} \text{tr}\left[(M + uu^\top) \Sigma_{-\kappa(i)}\right]\right)$$

Taking the logarithm and using $\log(1 + x) = x + \mathcal{O}(x^2)$:

$$\log \mu_i = \log p(y_i \mid \hat{\theta}_{-\kappa(i)}) + \frac{1}{2} \text{tr}\left[\left(\nabla_\theta^2 \log p(y_i \mid \hat{\theta}_{-\kappa(i)}) + \nabla_\theta \log p(y_i \mid \hat{\theta}_{-\kappa(i)}) \nabla_\theta \log p(y_i \mid \hat{\theta}_{-\kappa(i)})^\top\right) \Sigma_{-\kappa(i)}\right] + o_p \left($$

Substituting $\Sigma_{-\kappa(i)} = \frac{1}{n(1-1/K)} I(\theta_0)^{-1}$ and evaluating the derivatives at the true parameter θ_0 completes the proof. □

4 The Asymptotic Variance σ^2_{ELPPD} of the SNIS Estimator

We now derive the exact asymptotic variance σ^2_{ELPPD} of the SNIS estimator under the MCMC CLT as $S \to \infty$ followed by $n \to \infty$.

Theorem 2 (Asymptotic Variance Scaling)**.** *Under the high-dimensional BvM framework (Assumption 1), if the proposal distribution $q(\theta) = p(\theta \mid y^{(n)})$ (the full posterior proposal), then as $S \to \infty$ followed by $n \to \infty$, the asymptotic variance σ^2_{ELPPD} of the SNIS estimator $\widehat{ELPPD}$ scales as:*

$$\sigma^2_{\text{ELPPD}} = \frac{2d(n)}{(1 - 1/K(n))^2} \gamma_{\text{mcmc}} + \mathcal{O}_p\left(\frac{d(n)^2}{n}\right) \tag{5}$$

where γ_{mcmc} is the MCMC efficiency factor of the Markov chain targeting the full posterior $p(\theta \mid y^{(n)})$.

Proof. By the multivariate Delta method, the asymptotic variance is $\sigma^2_{\text{ELPPD}} = v^\top \Sigma v$. The gradient vector $v \in \mathbb{R}^{2n}$ is defined by $v_i = 1/\mathbb{E}_q[u_i] = 1/\mu_i$ and $v_{n+i} = -1/\mathbb{E}_q[w_{\kappa(i)}] = -1$. We consider the linear combination:

$$v^\top (Z^{(s)} - \mu_Z) = \sum_{i=1}^n \frac{w_{\kappa(i)}(\theta^{(s)})}{\mathbb{E}_q[w_{\kappa(i)}]} \left(\frac{p(y_i \mid \theta^{(s)})}{\mu_i} - 1\right) =: \sum_{i=1}^n h_i(\theta^{(s)})$$

4.1 Step-by-Step Expansion of the BvM Weight Ratio

By definition, the expectation of the unnormalized weight $w_k(\theta) = \frac{p(\theta|y_{\mathcal{D}_{-k}})}{q(\theta)}$ under the proposal distribution $q(\theta) = p(\theta \mid y^{(n)})$ is exactly 1:

$$\mathbb{E}_q\big[w_k(\theta)\big] = \int \frac{p(\theta \mid y_{\mathcal{D}_{-k}})}{p(\theta \mid y^{(n)})} p(\theta \mid y^{(n)})\, d\theta = \int p(\theta \mid y_{\mathcal{D}_{-k}})\, d\theta = 1$$

Therefore, $\frac{w_k(\theta)}{\mathbb{E}_q[w_k]} = w_k(\theta)$ holds exactly.

Under Assumption 1 (BvM), the fold-specific and full posteriors are approximated as:

$$p(\theta \mid y_{\mathcal{D}_{-k}}) = (2\pi)^{-d/2}|\Sigma_{-k}|^{-1/2} \exp\left(-\frac{1}{2}(\theta - \hat{\theta}_{-k})^\top \Sigma_{-k}^{-1}(\theta - \hat{\theta}_{-k})\right)$$

$$p(\theta \mid y^{(n)}) = (2\pi)^{-d/2}|\Sigma_n|^{-1/2} \exp\left(-\frac{1}{2}(\theta - \hat{\theta}_n)^\top \Sigma_n^{-1}(\theta - \hat{\theta}_n)\right)$$

where $\Sigma_n^{-1} = nI(\theta_0)$ and $\Sigma_{-k}^{-1} = n(1 - 1/K)I(\theta_0)$.

Let $\delta\theta = \theta - \hat{\theta}_n$ and $\Delta_k\theta = \hat{\theta}_n - \hat{\theta}_{-k}$. Substituting $\theta - \hat{\theta}_{-k} = \delta\theta + \Delta_k\theta$ into the ratio yields:

$$w_k(\theta) = \sqrt{\frac{|\Sigma_n|}{|\Sigma_{-k}|}} \exp\left(-\frac{1}{2}(\delta\theta + \Delta_k\theta)^\top \Sigma_{-k}^{-1}(\delta\theta + \Delta_k\theta) + \frac{1}{2}\delta\theta^\top \Sigma_n^{-1}\delta\theta\right)$$

Expanding the quadratic form in the exponent:

$$-\frac{1}{2}(\delta\theta + \Delta_k\theta)^\top [n(1 - 1/K)I(\theta_0)](\delta\theta + \Delta_k\theta) + \frac{1}{2}\delta\theta^\top [nI(\theta_0)]\delta\theta$$

$$= -\frac{1}{2}n(1 - 1/K)\delta\theta^\top I(\theta_0)\delta\theta - n(1 - 1/K)\delta\theta^\top I(\theta_0)\Delta_k\theta - \frac{1}{2}\Delta_k\theta^\top \Sigma_{-k}^{-1}\Delta_k\theta + \frac{1}{2}n\delta\theta^\top I(\theta_0)\delta\theta$$

Grouping the first and last terms gives:

$$-\frac{1}{2}n(1 - 1/K)\delta\theta^\top I(\theta_0)\delta\theta + \frac{1}{2}n\delta\theta^\top I(\theta_0)\delta\theta = \frac{n}{2K}\delta\theta^\top I(\theta_0)\delta\theta$$

Thus, we can write the weight ratio as:

$$w_k(\theta) = C_k \exp\left(-n(1 - 1/K)\delta\theta^\top I(\theta_0)\Delta_k\theta + \frac{n}{2K}\delta\theta^\top I(\theta_0)\delta\theta\right)$$

where $C_k = \sqrt{\frac{|\Sigma_n|}{|\Sigma_{-k}|}} \exp\left(-\frac{1}{2}\Delta_k\theta^\top \Sigma_{-k}^{-1}\Delta_k\theta\right)$.

Since the MCMC draws satisfy $\delta\theta = \mathcal{O}_p(1/\sqrt{n})$ and the training MLE discrepancy is $\Delta_k\theta = \mathcal{O}_p(1/n)$, the exponent is small. Taylor-expanding $\exp(x) \approx 1 + x$ yields:

$$w_k(\theta) \approx C_k \left(1 - n(1 - 1/K)\delta\theta^\top I(\theta_0)\Delta_k\theta + \frac{n}{2K}\delta\theta^\top I(\theta_0)\delta\theta\right)$$

Evaluating C_k using the determinant ratio:

$$\sqrt{\frac{|\Sigma_n|}{|\Sigma_{-k}|}} = \left(1 - \frac{1}{K}\right)^{-d/2} = \exp\left(-\frac{d}{2}\log(1 - 1/K)\right) \approx \exp\left(\frac{d}{2K}\right)$$

$$C_k \approx \exp\left(\frac{d}{2K} - \frac{1}{2}\Delta_k\theta^\top\Sigma_{-k}^{-1}\Delta_k\theta\right) \approx 1 + \frac{d}{2K} - \frac{1}{2}\Delta_k\theta^\top\Sigma_{-k}^{-1}\Delta_k\theta$$

Multiplying the terms:

$$w_k(\theta) \approx 1 - n(1-1/K)\delta\theta^\top I(\theta_0)\Delta_k\theta + \frac{n}{2K}\delta\theta^\top I(\theta_0)\delta\theta + \text{const. terms}$$

where the constant terms (independent of MCMC draw θ) are $\frac{d}{2K} - \frac{1}{2}\Delta_k\theta^\top\Sigma_{-k}^{-1}\Delta_k\theta$. Since MCMC fluctuations are generated only by terms depending on $\delta\theta$, these constants do not affect the asymptotic variance.

4.2 Step-by-Step Expansion of the Likelihood Ratio

We Taylor-expand $p(y_i \mid \theta)$ around the full-data MLE $\hat{\theta}_n$:

$$p(y_i \mid \theta) = p(y_i \mid \hat{\theta}_n)\left(1 + \nabla_i^\top\delta\theta + \frac{1}{2}\delta\theta^\top(\nabla_i^2 + \nabla_i\nabla_i^\top)\delta\theta + o_p(d/n)\right)$$

where $\nabla_i = \nabla_\theta \log p(y_i \mid \hat{\theta}_n)$ and $\nabla_i^2 = \nabla_\theta^2 \log p(y_i \mid \hat{\theta}_n)$.

Integrating the joint expansion of $p(y_i \mid \theta)w_k(\theta)$ under $\delta\theta \sim \mathcal{N}(0, \frac{1}{n}I(\theta_0)^{-1})$ yields:

$$\mu_i \approx p(y_i \mid \hat{\theta}_n)\left(1 + \frac{1}{2n}\text{tr}\left((\nabla_i^2 + \nabla_i\nabla_i^\top)I(\theta_0)^{-1}\right) + \frac{d}{K} - (1-1/K)\nabla_i^\top\Delta_k\theta\right)$$

Taking the ratio $\frac{p(y_i|\theta)}{\mu_i}$ and expanding using $(1+x)^{-1} \approx 1 - x$:

$$\frac{p(y_i \mid \theta)}{\mu_i} - 1 \approx \nabla_i^\top\delta\theta + \frac{1}{2}\left[\delta\theta^\top(\nabla_i^2 + \nabla_i\nabla_i^\top)\delta\theta - \frac{1}{n}\text{tr}\left((\nabla_i^2 + \nabla_i\nabla_i^\top)I(\theta_0)^{-1}\right)\right] - \frac{d}{K} + (1-1/K)\nabla_i^\top\Delta_{\kappa(i)}\theta$$

4.3 Evaluation of the Linear Combination and First-Order Cancellation

We multiply the expansions of $w_{\kappa(i)}(\theta)$ and $\frac{p(y_i|\theta)}{\mu_i} - 1$, then sum over $i = 1, \ldots, n$:

$$\sum_{i=1}^n h_i(\theta) \approx \sum_{i=1}^n \nabla_i^\top\delta\theta + F(\theta) + o_p(1/n)$$

Since $\hat{\theta}_n$ is the MLE of the full dataset, the score equations are satisfied exactly: $\sum_{i=1}^n \nabla_i = 0$. This implies:

$$\sum_{i=1}^n \nabla_i^\top\delta\theta = \left(\sum_{i=1}^n \nabla_i\right)^\top \delta\theta = 0 \quad \text{(exactly)}$$

Thus, the first-order MCMC fluctuation term vanishes. Collecting the second-order MCMC fluctuations:

$$F(\theta) \approx \sum_{i=1}^n \left[\frac{1}{2}\delta\theta^\top(\nabla_i^2 + \nabla_i\nabla_i^\top)\delta\theta - n(1-1/K)(\delta\theta^\top I(\theta_0)\Delta_{\kappa(i)}\theta)(\nabla_i^\top\delta\theta)\right]$$

- **Hessian term**: Since $\sum_{i=1}^{n}(\nabla_i^2+\nabla_i\nabla_i^\top) = \mathcal{O}_p(\sqrt{nd})$, the term $\sum_i \frac{1}{2}\delta\theta^\top(\nabla_i^2+\nabla_i\nabla_i^\top)\delta\theta = \mathcal{O}_p(\sqrt{d}/n)$ is asymptotically negligible.

- **Dominant term**:

$$-n(1-1/K)\delta\theta^\top I(\theta_0)\left(\sum_{i=1}^{n}\Delta_{\kappa(i)}\theta\nabla_i^\top\right)\delta\theta$$

By partitioning the sum over the K folds, we have $\sum_{i=1}^{n}\Delta_{\kappa(i)}\theta\nabla_i^\top = \sum_{k=1}^{K}\Delta_k\theta G_k^\top$, where $G_k = \sum_{j\in\mathcal{D}_k}\nabla_j$. Using the training MLE expansion:

$$\Delta_k\theta = \hat{\theta}_n - \hat{\theta}_{-k} = \frac{K}{n(K-1)}I(\theta_0)^{-1}G_k + \mathcal{O}_p\left(\frac{\sqrt{d}}{n^{3/2}}\right)$$

Substituting this back:

$$\sum_{k=1}^{K}\Delta_k\theta G_k^\top = \frac{K}{n(K-1)}I(\theta_0)^{-1}\sum_{k=1}^{K}G_kG_k^\top + \mathcal{O}_p\left(\frac{\sqrt{d}}{n}\right)$$

Since G_k has covariance $\mathrm{Var}(G_k) = \frac{n}{K}I(\theta_0)$, the sum of outer products is $\sum_k G_kG_k^\top = nI(\theta_0) + \mathcal{O}_p(\sqrt{nd})$. Therefore:

$$\sum_{i=1}^{n}\Delta_{\kappa(i)}\theta\nabla_i^\top = \frac{K}{K-1}I_d + \mathcal{O}_p\left(\frac{\sqrt{d}}{n}\right)$$

Multiplying by $-n(1-1/K) = -n\frac{K-1}{K}$ yields the simplified leading-order term:

$$F(\theta^{(s)}) \approx -n\delta\theta^{(s)\top}I(\theta_0)\delta\theta^{(s)} = -\xi^{(s)\top}I(\theta_0)\xi^{(s)}$$

4.4 MCMC Autocovariance and Wick's Theorem Derivation

The stationary distribution of the standardized draws $\xi^{(s)} = \sqrt{n}(\theta^{(s)} - \hat{\theta}_n)$ is $\mathcal{N}(0, \Sigma_{\text{post}})$, where the training set posterior covariance is scaled: $\Sigma_{\text{post}} = \frac{1}{1-1/K}I(\theta_0)^{-1}$. We write $\xi^{(s)} = \sqrt{\frac{1}{1-1/K}}I(\theta_0)^{-1/2}x^{(s)}$, where $x^{(s)} \sim \mathcal{N}(0, I_{d(n)})$ is a stationary standardized Markov chain. The quadratic form is:

$$F(\theta^{(s)}) = -\frac{1}{1-1/K}x^{(s)\top}x^{(s)} = -\frac{1}{1-1/K}Q^{(s)}$$

The MCMC asymptotic variance of $Q^{(s)} = x^{(s)\top}x^{(s)}$ is derived by modeling $x^{(s)}$ as a stationary Gaussian AR(1) process with correlation ρ:

$$x^{(s)} = \rho x^{(s-1)} + \sqrt{1-\rho^2}\epsilon^{(s)}$$

The sum $Q^{(s)} = \sum_{a=1}^{d(n)} (x_a^{(s)})^2$ is a sum of $d(n)$ independent AR(1) components. We compute the autocovariance using Isserlis/Wick's theorem:

$$\mathrm{Cov}\left((x_a^{(0)})^2, (x_a^{(l)})^2\right) = \mathbb{E}\left[(x_a^{(0)})^2 (x_a^{(l)})^2\right] - \mathbb{E}\left[(x_a^{(0)})^2\right] \mathbb{E}\left[(x_a^{(l)})^2\right]$$

Since $x_a^{(0)}, x_a^{(l)}$ are jointly Gaussian with mean zero, variance 1, and correlation ρ^l:

$$\mathbb{E}\left[(x_a^{(0)})^2 (x_a^{(l)})^2\right] = \mathbb{E}[(x_a^{(0)})^2]\mathbb{E}[(x_a^{(l)})^2] + 2\left(\mathbb{E}[x_a^{(0)} x_a^{(l)}]\right)^2 = 1 \times 1 + 2(\rho^l)^2 = 1 + 2\rho^{2l}$$

Subtracting the product of the expectations ($1 \times 1 = 1$) yields:

$$\mathrm{Cov}\left((x_a^{(0)})^2, (x_a^{(l)})^2\right) = 2\rho^{2l}$$

Summing the geometric series to obtain the spectral density at zero yields:

$$\sigma_Q^2 = \mathrm{Var}(Q^{(0)}) + 2\sum_{l=1}^{\infty} \mathrm{Cov}(Q^{(0)}, Q^{(l)}) = 2d(n) + 4d(n)\sum_{l=1}^{\infty} \rho^{2l} = 2d(n)\left(1 + \frac{2\rho^2}{1-\rho^2}\right) = 2d(n)\gamma_{\mathrm{mcmc}}$$

where $\gamma_{\mathrm{mcmc}} = \frac{1+\rho^2}{1-\rho^2}$ is the mathematically defined MCMC efficiency factor [5]. Therefore:

$$\sigma^2_{\mathrm{ELPPD}} = \mathrm{Var}_{\mathrm{mcmc}}(F(\theta^{(s)})) = \frac{2d(n)}{(1 - 1/K(n))^2}\gamma_{\mathrm{mcmc}} + \mathcal{O}_p\left(\frac{d(n)^2}{n}\right)$$

This completes the proof. □

5 Bias of the Asymptotic Variance Estimator $\widehat{\sigma}^2$

We characterize the bias of the MCMC variance estimator $\widehat{\sigma}^2$ under the frequentist data-generating distribution in the limit $S \to \infty$ followed by $n \to \infty$.

Theorem 3 (Asymptotic Bias as a Function of K and n)**.** *Let $\widehat{\sigma}^2$ be the consistent estimator of the MCMC asymptotic variance $\sigma^2_{\mathrm{ELPPD}}$ as $S \to \infty$. Under the data-generating distribution P_{θ_0}, the bias of $\widehat{\sigma}^2$ relative to the true frequentist variance of the cross-validation estimator* $\mathrm{Var}_{Y^{(n)}}(\mathrm{ELPPD})$ *is:*

$$\mathit{Bias}\big(\widehat{\sigma}^2\big) := \mathbb{E}_{Y^{(n)}}\left[\sigma^2_{\mathrm{ELPPD}}\right] - \mathrm{Var}_{Y^{(n)}}(\mathrm{ELPPD}) = \frac{2d(n)\gamma_{\mathrm{mcmc}}}{(1 - 1/K(n))^2} - \mathrm{Var}_{Y^{(n)}}(\mathrm{ELPPD}) + \mathcal{O}\left(\frac{d(n)^2}{n}\right) \tag{6}$$

Proof. As $S \to \infty$, the estimator $\widehat{\sigma}^2$ converges in probability to $\sigma^2_{\mathrm{ELPPD}}$. The expectation under the data-generating distribution is:

$$\mathbb{E}_{Y^{(n)}}\left[\sigma^2_{\mathrm{ELPPD}}\right] = \frac{2d(n)\gamma_{\mathrm{mcmc}}}{(1 - 1/K(n))^2} + \mathbb{E}_{Y^{(n)}}\left[\mathcal{O}_p\left(\frac{d(n)^2}{n}\right)\right] = \frac{2d(n)\gamma_{\mathrm{mcmc}}}{(1 - 1/K(n))^2} + \mathcal{O}\left(\frac{d(n)^2}{n}\right)$$

Subtracting the true variance of the cross-validation estimator $\mathrm{Var}_{Y^{(n)}}(\mathrm{ELPPD})$ under the data-generating distribution yields the stated bias formula. □

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